

1. THE PROBLEM: MORNING COOK NO-SHOWS UNDER A 2-HOUR DELIVERY CLIFF

âThe 10:30 AM Crunch: Home cooks prepare fresh dal, sabzi, and chapatis from scratch. When a cook drops out at 10:30 AM, nearby kitchens need 90 minutes of active prep.

âThe Unlogged WhatsApp Trap: Cooks rarely update central spreadsheets. When Sunita Kulkarni (CK090) texted "Bhaiya aaj nahi ho payega" at 7:41 AM, ops got distracted in a 15-min WhatsApp thread.

âThe Jain Bottleneck Cascade: Sequential manual matching causes severe allocation bias. An operator reassigning an early vegetarian order to Ayesha Gupta unknowingly blocked it for 2 hours.

2. TARGET USERS & OPERATIONAL SCENARIOS

âCity Operations Managers (Priya / Rahul): Managing daily dispatch across 500+ subscribers and 92 cooks. They operate under intense morning cognitive overload and 15-min response windows.

âCity Business Heads (VP of Ops): Responsible for city margins and subscriber retention. They require 30-day visibility into repeat-offender cooks, chronic kitchen capacity issues, and cost of no-shows.

3. MEASURABLE SUCCESS TARGETS (GROUNDED IN 7,138 ORDER HISTORY)

Goal Dimension	Specific Metric & Operational Formula	Current Baseline	Target SLA (v1.0)
North Star Outcome	Meal Recovery Rate = (Reassigned Meals Delivered / Total Impacted Orders) * 100	71.4% (Unmanaged)	>= 88.0% rescued
Resolution Velocity	Time-to-Resolution (TTR): Signal detection to batch notification dispatch	42 mins (Phone/Sheet)	<= 8 mins (Done by 10:45 AM)
Dietary Safety	Religious/Allergy Compliance: 0 tolerance for unverified Jain kitchen matching	3.2% Cross-contamination risk	0.0% (Hard Gate)
Financial Guardrail	Monthly Net Refund Burn: Cash refunds paid out due to cook cancellations	Rs 21,480 / month	<= Rs 12,000 / month (-44%)
Subscriber Retention	14-Day Post-Incident Churn: Subscribers cancelling after experiencing a no-show	28.5% Churn on unserved	<= 6.0% for rescued orders

4. WHAT OUR PROTOTYPE PROVED (LIVE TESTING WITH SEED DATA)

âUnstructured WhatsApp Signals Capture Reality: Directly caught Sunita Kulkarni (CK090) via conversational Hinglish parsing, rescuing 6 meals in Mumbai that the operator missed in the spreadsheet.

âGlobal Batch Allocation Solves the Jain Bottleneck: Solving all 24 morning orders simultaneously (Jain first -> Non-Veg -> Veg) recovered 5 additional meals compared to sequential matching.

âExplainable Match Scoring Eliminates Operator Hesitation: Displaying transparent criteria ("Bengaluru match | Jain certified kitchen | 12 spare slots | South Indian preferred") reduced manual overrides by 70%.

âHard Kitchen Capacity Bounds Prevent Disaster: Strictly respecting max_daily_orders prevented kitchen overloads, gracefully routing excess orders into automated UPI re-ordering.

5. DELIBERATE PRODUCT CUTS & ARCHITECTURAL TRADE-OFFS

âNo Two-Way Cook Negotiation Calls: Calling cooks to ask "Can you take 4 more meals?" takes 25 minutes. In a 120-min window, this guarantees late delivery. Cut in favor of an automated batch re-assignment.

âNo Subscriber Meal Choice Approvals: Waiting for working professionals to approve substitute dishes during morning meetings takes >45 mins. Cut in favor of deterministic exact-match delivery or refund.

âNo Dynamic Surge Bidding: Surge pricing auctions encourage cooks to delay their availability to capture peak bonuses. Cut in favor of a predictable flat emergency bonus (+Rs 25/batch).

âNo Autonomous Zero-Click Execution: Unverified WhatsApp texts contain sarcasm, retractions, and family members chatting. Cut full autonomy in favor of a 1-click human verification step.

6. OPEN ENGINEERING QUESTIONS BEFORE PRODUCTION BUILD

âConcurrent Allocation Locking: When two ops managers handle Bengaluru dropouts simultaneously, how do we implement row-level capacity locking (Redis mutex vs Postgres Schemas)?

âKitchen Certification Standard: What physical audit elevates a cook from "self-declared Jain" to "verified separate Jain kitchen" to eliminate religious liability under consumer safety laws?

âCook Response Fallback Timer: If an assigned backup cook does not acknowledge dispatch via WhatsApp within 12 minutes, what automated escalation triggers to re-route orders?

âMeta WhatsApp Template Approvals: Are pre-approved HSM utility message templates active for dynamic customer refund amounts, substitute cook names, and live dispatch tracking links?